Verifying test case for smooth deformation energy density

Testcase description: Plunger Transformer with 5 rectangular elements in x-axis, 3 elements in y-axis. Element length $l = 25\mu m$. Based on Mechanical displacement analysis. Mechanical displacement is linear $[0, 1.50709 \cdot 10^{-8}]$. No displacement in y-axis. Poisson number $\nu = 0.49$. Elasticity modulus E = 1 $\frac{N}{m^2}$
smooth deformation energy density is calculated as follows:

$$\text{elasticity tensor } C = E\frac{1-\nu}{(1+\nu)(1-2\nu)}\begin{bmatrix} 1 & \frac{\nu}{1-\nu} & 0 \\ \frac{\nu}{1-\nu} & 1 & 0 \\ 0 & 0 & \frac{1-2\nu}{2(1-\nu)} \end{bmatrix} = \begin{bmatrix} 17.1141 & 16.443 & 0 \\ 16.443 & 17.1141 & 0 \\ 0 & 0 & 0.3356 \end{bmatrix}$$

$$\text{strain tensor was simplified because only displacement in x-axis: } \epsilon = \begin{bmatrix} \frac{du}{dx} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1.2057 \cdot 10^{-4} \\ 0 \\ 0 \end{bmatrix}$$

$$\text{smooth deformation energy density: } w = \frac{1}{2}\epsilon^T C \epsilon = 1.244 \cdot 10^{-7}\frac{J}{m^3}$$